

Lab Exercises for 2.086 Week 11

(Spring 2025)

Problem 1: Unsupervised Learning & *k*-Means Clustering

Move from classification to **clustering**, code *k*-means from scratch, and use internal metrics (SSE & silhouette) to choose an appropriate number of clusters for a simple bicycle-geometry dataset.



Load and Inspect the Data

The dataset contains eight geometry measurements per bicycle and a **reference label** (1 = Road, 2 = Mountain). Labels are *not* used for clustering—they let us sanity-check results later.

```
clear all; close all;
rng(0); % reproducibility
fileName = 'Bikestyle_y.csv';

T = readtable(fileName, 'PreserveVariableNames', true);
featIdx2 = [1 6]; % two-feature subset
featNames2 = T.Properties.VariableNames(featIdx2);

X2 = T{:, featIdx2}; % N×2 matrix
Xall = T{:, 1:end-1}; % N×8 matrix (exclude label column)
y = T{:, end}; % ground-truth labels (not used for clustering)

%Labels serve as a reference for evaluation at the end.
```

Two-Feature Experiment

We start simple with only **two geometry features** so the clusters can be visualised directly.

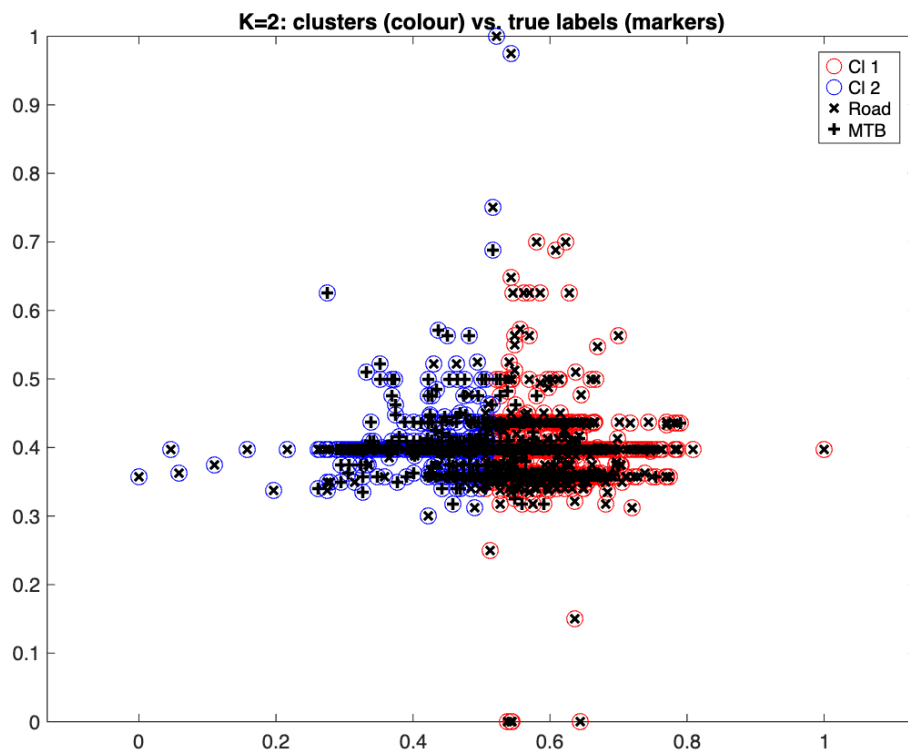
Min-max scaling keeps both features on the same [0, 1] range.

```
% ---- Scaling ----  
min2    = min(X2);  
range2  = max(X2) - min2;  range2(range2==0)=1;  
X2mm    = (X2 - min2) ./ range2;
```

Run *k*-Means for K = 2–4

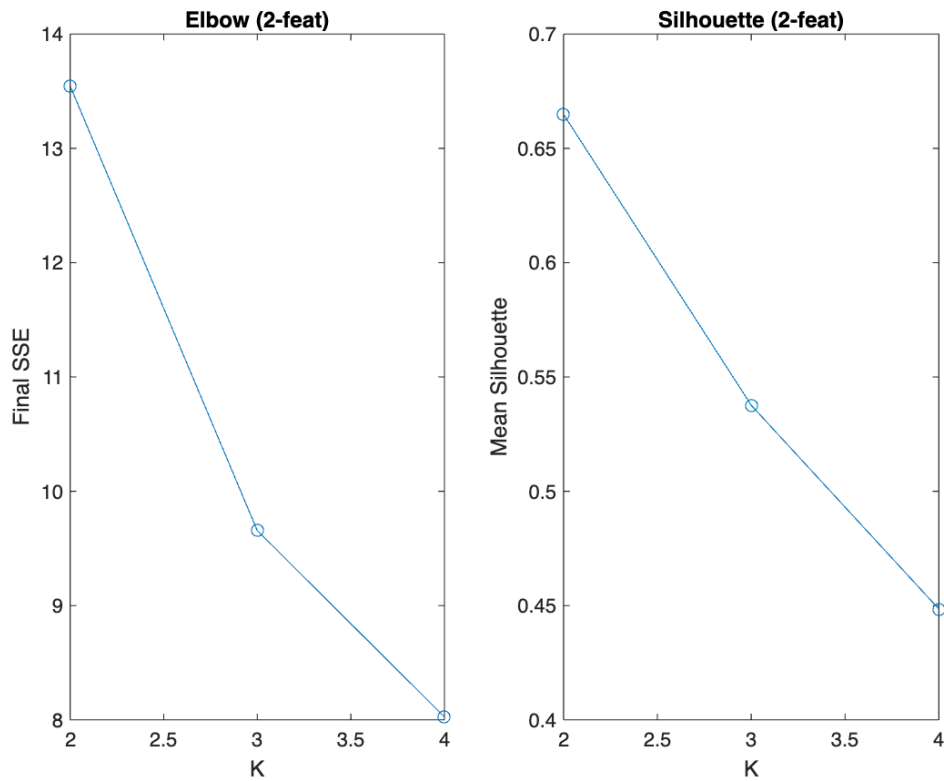
We record the **final SSE** and the **mean silhouette** for each K.

```
Ks    = 2:4;  SSEs = zeros(size(Ks)); MS = zeros(size(Ks));  
for i=1:numel(Ks)  
    K = Ks(i);  
    [idx,C,SSE] = myKMeans(X2mm,K);  
    SSEs(i)=SSE(end);  
    MS(i)=meanSilhouette(X2mm,idx);  
  
    if K==2          % nice visual for the K=2 case  
        figure;  
        gscatter(X2mm(:,1),X2mm(:,2),idx,'rb','oo',8,'off'); hold on;  
        plot(X2mm(y==1,1),X2mm(y==1,2),'kx','MarkerSize',6,'LineWidth',1.5);  
        plot(X2mm(y==2,1),X2mm(y==2,2),'k+','MarkerSize',6,'LineWidth',1.5);  
        legend({'Cl 1','Cl 2','Road','MTB'}); axis equal;  
        title('K=2: clusters (colour) vs. true labels (markers)');  
    end  
end
```



Diagnostics.

```
figure;
subplot(1,2,1); plot(Ks,SSEs,'-o'); xlabel('K'); ylabel('Final SSE');
title('Elbow (2-feat)');
subplot(1,2,2); plot(Ks,MS,'-o'); xlabel('K'); ylabel('Mean Silhouette');
title('Silhouette (2-feat)');
```



We observe that K=2 has a high score.

Eight-Feature Experiment (K = 2)

With all eight geometry features the clusters cannot be visualised easily, but SSE and silhouette still tell us how compact/coherent they are.

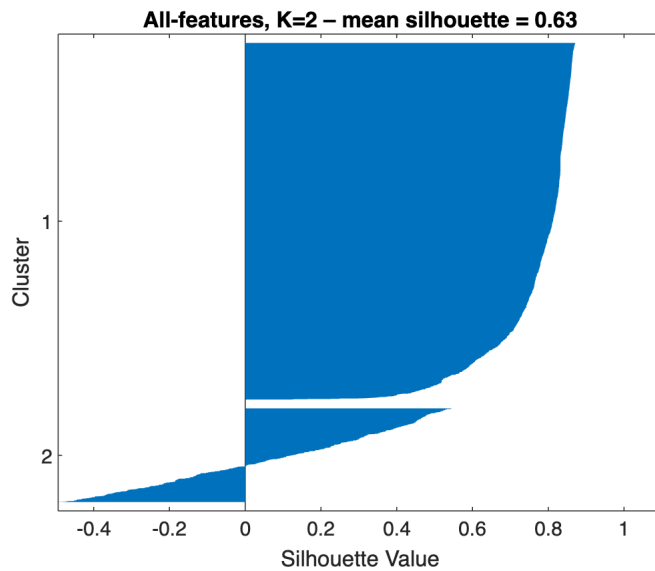
4.1 Handle Missing Values & Scale

```
XallImp = fillmissing(Xall,'movmean',size(Xall,1)); % simple mean-impute
minA = min(XallImp,[],'omitnan');
rangeA = max(XallImp,[],'omitnan')-minA; rangeA(rangeA==0)=1;
Xallmm = (XallImp-minA)./rangeA;

[idxAll,CAll,SSEAll] = myKMeans(Xallmm,2);
mscAll = meanSilhouette(Xallmm,idxAll);
fprintf('All-features → SSE = %.2f | mean silhouette = %.2f\n',SSEAll(end),mscAll);
```

```
All-features → SSE = 51.63 | mean silhouette = 0.63
```

```
figure; silhouette(Xallmm,idxAll);
title(sprintf('All-features, K=2 - mean silhouette = %.2f',mscAll));
```



If the mean silhouette decreases after adding the extra six features, it usually indicates that some of those dimensions add noise or blur the separation that was crisp in the 2-D. Distances in high-dimensional space become harder to interpret (the so-called **curse of dimensionality**), and irrelevant or weakly informative features can pull points toward the wrong centroid.